

SERGEY ORLOV

SENIOR DATA SCIENTIST | PRODUCTION ML | DATA ENGINEERING

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PROFILE

Data scientist and technical leader with a PhD in Physics and Mathematics and experience spanning quantitative modeling, applied machine learning, and production data systems. Hands-on ownership across problem definition, model development, software implementation, deployment, and ongoing maintenance.

PROFESSIONAL EXPERIENCE

Data Scientist | Cashco Financial

April 2024 - Present | Calgary, Canada

- Design, build, and maintain credit decisioning software, connecting analytical work with production engineering and ongoing operation.
- Develop and evaluate machine-learning models and support the implementation of analytical methods in business workflows.
- Build document-processing and data-preparation workflows, with attention to input quality and the usability of structured outputs.
- Develop analytics and reporting infrastructure spanning ingestion, transformation, analytical modeling, and business reporting.
- Collaborate with business stakeholders to translate requirements into maintainable software and communicate technical decisions.
- Take responsibility for implementation quality, evaluation, documentation, and ongoing improvements.

PUBLIC ENGINEERING WORK

Independent demonstrations make my engineering approach inspectable. Synthetic labs are separate from employer systems and results.

Open Decisioning Lab - Synthetic model evaluation, calibration, API contracts, and explicit decision policy.

Default Risk Lifecycle Lab - Chronological evaluation, label maturity, drift diagnostics, and monitoring.

Document AI Reliability Lab - Synthetic extraction evaluation and quality gates.

Cashflow Intelligence Lab - Synthetic transaction normalization and feature engineering.

Analytics Engineering Examples - Generic ingestion, dbt modeling, and platform design discussions.

EXPERIENCE & TECHNICAL FOUNDATION

Data Scientist | Cybera Inc.

October 2023 - March 2024 | Calgary, Canada

- Designed and deployed an entity-resolution pipeline for unstructured data using learned representations, retrieval, and machine-learning methods.
- Worked on scalable execution using containerization, multiprocessing, cloud infrastructure, and GPU acceleration.

Director of Data Science & HPC | Tomsk State University

May 2016 - December 2022 | Tomsk, Russia

- Established and led a data science laboratory, translating research into applied services and mentoring students in machine learning and software engineering.
- Managed engineering work for shared computing infrastructure and guided technical design, operations, and delivery.

Computational Modelling & ML Researcher | Tomsk State University

February 2012 - May 2016 | Tomsk, Russia

- Developed scientific simulation and predictive-modeling software using C++ and parallel computing.
- Applied machine learning and optimization methods to scientific problems and improved Linux/HPC workflows.

TECHNICAL SKILLS

Modeling: Python, SQL, XGBoost, LightGBM, CatBoost, scikit-learn, TensorFlow, calibration, explainability, evaluation, monitoring.

Engineering: FastAPI, Docker, Git, Linux, CI/CD, REST APIs, testing, Azure, GCP, AWS.

Data: Airflow, dbt, PySpark, Airbyte, PostgreSQL, MS SQL, MySQL, ETL/ELT, analytical modeling, Metabase, Power BI, Tableau.

Applied AI: Document processing, transformer embeddings, retrieval, FAISS, vision-language models, quality evaluation.

EDUCATION

Master of Data Science and Analytics

University of Calgary | 2023

PhD, Physical and Mathematical Science

Tomsk State University | 2012